

Deepak Mallampati

APPLIED AI ENGINEER

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PROFESSIONAL SUMMARY

Applied AI Engineer with 4+ years building production-grade agentic LLM and RAG systems for regulated enterprises across healthcare, banking, and financial services. Architect of conductor-subagent orchestration frameworks and HIPAA-compliant AI workflows with RBAC, audit logging, and data governance built in. Daily Claude Code practitioner - built **career-ops**, an autonomous AI pipeline, on it. Track record of measurable outcomes: 42% token reduction, 40% hallucination reduction, 60% fewer failed production runs, and SLA-grade reliability at enterprise scale.

CORE COMPETENCIES

Agentic AI & Multi-Agent Systems • MCP / Tool-Use Integration • Claude Code & AI-Assisted Development • HIPAA / RBAC / Data Governance • RAG Architecture • Python & TypeScript • REST APIs / OAuth2 / Webhooks • Enterprise Workflow Automation

Generative AI & LLMs: Claude Sonnet, GPT-4o/5, Gemini, LLaMA | LangChain/LangGraph | RAG | Multi-Agent Systems | Fine-tuning (LoRA, QLoRA, PEFT) | Prompt Engineering

Agentic & Integration: Conductor-subagent orchestration | MCP/tool-use | REST APIs & microservices | OAuth 2.0, JWT, webhooks | iPaaS-style batch orchestration

Security & Compliance: HIPAA, SOC 2, PCI DSS | RBAC | PII Masking/Redaction | Audit Logging | Differential Privacy, k-anonymity, l-diversity

Data & Cloud: PostgreSQL, MongoDB, Redis | SQL & data modeling | AWS (Bedrock, Lambda, S3), Azure OpenAI, GCP Vertex | Docker, Kubernetes, Terraform

Languages & Tools: Python, TypeScript/JavaScript, Java, SQL | FastAPI, React | MLflow, W&B | RAGAS, BERTScore

PROFESSIONAL EXPERIENCE

AI Engineer - Progress Solutions

Jul 2025 - Present | USA

- Architected production-grade **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex multi-step objectives into context-scoped subagents that execute autonomously with reliable, auditable outcomes - cutting token consumption **42%**.
- Designed **HIPAA-compliant data workflows** over sensitive clinical datasets with RBAC, audit logging, and privacy guarantees (k-anonymity, l-diversity, differential privacy) - enabling compliant analytics with zero PII exposure.
- Engineered **RAG pipelines** over large healthcare document corpora (dense retrieval + cross-encoder re-ranking) to lift answer relevance and materially reduce hallucination.
- Built fault-tolerant **automation infrastructure** (scheduled batch orchestration + exponential-backoff retries), reducing failed production runs **60%**.
- Established an **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards, defining success metrics and surfacing regressions before release.

AI Engineer Intern - Vanguard

Jan 2024 - Jan 2025 | USA

- Embedded with product, platform, and data teams to ship **25+ AI agents** on Vanguard's internal AI platform - translating business workflow gaps into AI-powered prototypes for 20+ internal teams.
- Built **12 APIs and microservices** (REST, OAuth2-secured) powering new AI capabilities, cutting p95 latency 1.5s → **800ms** and accelerating integration **40%**.
- Engineered content safeguards and policy-compliant controls, reducing unsafe outputs **42%**; evaluated GPT-4o/Claude/Gemini, improving task success **27%**.
- Supported **100,000+ requests daily** across internal teams with SLA-grade reliability.

ML & Gen AI Research Assistant - Kent State University

Oct 2023 - Jan 2024 | USA

- Built a privacy-preserving ML pipeline on clinical hospital-readmission data, cutting re-identification risk **3.38%** → **0.32%** while retaining **99%** of baseline predictive performance.
- Designed a composite privacy-utility scoring framework benchmarking six anonymization configurations to identify the optimal operating point.

ML Trainee - Emerson

Jun 2022 - Apr 2023 | India

- Built supervised regression/classification models for equipment-failure prediction and anomaly detection; fine-tuned BERT/RoBERTa for NER, achieving **89% F1** on domain tasks.

PROJECTS

career-ops

An autonomous, Claude-Code-native AI pipeline that scans listings, scores fit, and generates tailored applications overnight without supervision - agentic orchestration applied to a real workflow.

Privacy-Preserving Clinical ML Pipeline

A HIPAA-aligned framework combining k-anonymity, l-diversity, and differential privacy to analyze sensitive patient data without exposing identities, quantifying the privacy-utility tradeoff.

MangaLens

A shipped Chrome extension (live on the Chrome Web Store) delivering real-time AI translation across 29+ sites - productionized LLM feature with real users.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 | GPA 3.70/4.0